

A Comparative Study for Detecting Polycystic Ovary Syndrome Using A Machine Learning Framework

Tamanna Hossain Mou¹, Oishi Jyoti², Tamal Ahmed³, and MD. Rafi Imam⁴

^{1,2,3,4}*Department of Electrical & Computer Engineering*

^{1,2,3,4}*Rajshahi University of Engineering & Technology*

Rajshahi-6204, Bangladesh

tamannamou128@gmail.com¹, oj.ruet@gmail.com², tamalahmed1144@gmail.com³, and rafimam96@gmail.com⁴

Abstract—Polycystic Ovary Syndrome (PCOS) is a widespread hormonal disorder affecting numerous women with ovaries, during their reproductive ages, often leading to various complications if undiagnosed or improperly managed. This research aims to improve PCOS detection by utilizing an interpretable machine-learning framework for diagnostic assistance. We utilized a publicly available dataset from Kaggle, featuring 541 samples and 44 attributes. Various preprocessing techniques and feature selection methods, such as Chi-Square, Mutual Information, and Correlation Coefficient, were employed here. Subsequently, we trained multiple machine learning algorithms and subjected them to K-fold cross-validation. Their performance is evaluated and compared by accuracy, precision, recall, and F1 score. Our findings reveal that the CatBoost algorithm, when paired with Chi-Square feature selection, achieved the highest accuracy rate of 93.9%. Our study focuses on investigating important features, comparing algorithm performance, and achieving high accuracy in PCOS detection. The research validates machine learning's effectiveness for non-invasive PCOS diagnosis, laying the groundwork for future medical diagnostics.

Index Terms—Polycystic Ovary Syndrome, CatBoost, Chi-Square, Feature Selection, Diagnosis, Comparative Analysis

I. INTRODUCTION

Polycystic Ovary Syndrome (PCOS) is a hormonal disorder that hinders the proper maturation of egg cells, preventing them from being adequately fertilized by sperm. This endocrine condition is prevalent in approximately 10-15% of women of reproductive age globally [1]. The World Health Organization estimates that around 117 million women globally, or roughly 3.5% of the female population, are affected by Polycystic Ovary Syndrome [2]. A common feature of PCOS is the overproduction of androgens by the ovaries, which are male hormones that are typically present in low concentrations in women. This hormonal imbalance can lead to irregular menstrual cycles and contribute to various symptoms associated with the condition [3]. PCOS manifests through various signs such as inconsistent menstruation, skin breakouts, heavy

menstrual flow, unusual hair growth, skin darkening and thickening, weight fluctuations, discomfort in the pelvic region, greasy skin, and fertility challenges. The condition is marked by elevated levels of male hormones, resistance to insulin, failure of the ovaries to release eggs during the menstrual cycle, and disruptions in the neuroendocrine system. Machine learning, which enables computers to automatically learn from their surroundings, is a crucial field in computer science. It plays a crucial part in modern healthcare systems [3]. Machine learning provides an efficient tool for accurately diagnosing Polycystic Ovary Syndrome (PCOS). Several research studies have explored the application of machine learning within the realm of Polycystic Ovary Syndrome (PCOS). Khoirun et al. [4] proposed the performance comparison by using a confusion matrix between random forest and logistic regression algorithms for detecting PCOS. The feature selection method and the cross-validation process were used. The best accuracy is 91%, which was achieved from the RF model. Amsy et al. [5] carried out Principal Component Analysis (PCA), using a variety of machine learning methods, including logistic regression, K-Nearest Neighbours (KNN), Naïve Bayes, Classification and Regression Trees (CART), Support Vector Machine (SVM), and Random Forest Classifier. The results suggest that the Random Forest Classifier stands out as the most precise approach in forecasting PCOS, demonstrating an accuracy rate of 89.02%. Vaidehi et al. [6] proposed the following five different machine learning models: K-Nearest Neighbours, Random Forest, Gaussian Naive Bayes, Support Vector Machine (SVM), and Logistic Regression. From the initial dataset of 41 features, the top 30 attributes were retrieved using the CHI-SQUARE approach. Upon assessing each model's performance, it was concluded that the Random Forest Classifier, with a rate of 90.9%, had the highest accuracy and reliability. Dana et al. [7] developed an efficient diagnostic model using MATLAB. They studied seven

different classifiers (KNN, Neural Network, Naïve Bayes, SVM, Classification Tree, Logistic Regression, and Linear Discriminant). By comparing their accuracies and results, they got 92.60% highest accuracy for LD.

The aim of this research paper is to employ a variety of preprocessing techniques for data enhancement, utilize data visualization methods, and investigate significant features correlated with Polycystic Ovary Syndrome (PCOS). Additionally, the study aims to apply multiple feature selection methods, along with cross-validation (CV), to enhance the model's robustness. Furthermore, the research seeks to compare the performance of various algorithms, evaluating accuracy, precision, recall, and F1 score as metrics. The ultimate goal is to achieve improved accuracy in PCOS detection through a comprehensive and systematic approach. The most remarkable result, with a 93.9% accuracy rate, has been attained by applying the Chi-square feature selection method to the CatBoost classifier in this study, surpassing the previous results. The remainder of the paper is structured as follows: Section 2 elaborates on the system's architecture; Section 3 discusses the results; and Section 4 provides the conclusion along with insights on future or planned work.

II. METHODOLOGY

This study has utilized a comparative approach involving various feature extraction techniques and computational algorithms, with Python serving as the programming language for machine learning implementation. A schematic representation of the proposed systems is depicted in Figure 1.

As indicated by Figure 1, the study follows multiple procedural steps. Initially, characteristics and attributes from the PCOS dataset are fed into the system. Then data processing, data balancing, scaling, and different feature selection methods are performed on the data. For finding significant features that are correlated with the target variable, three distinct feature selection methods (Chi-Square, Mutual Information, Correlation Coefficient) are applied. Subsequently, the refined data underwent cross-validation. Once prepared, the selected data are subjected to training through various machine learning models like Random Forest, SVM, KNN, Logistic Regression, Linear Discriminant, LightGBM, CatBoost and XGBoost. Finally, the effectiveness of different feature selection techniques and the accuracy of different algorithms are evaluated and compared.

A. Dataset Description

The data is sourced from Kaggle. This specific dataset, named "Polycystic Ovary Syndrome," was gathered from 10 hospitals in Kerala, India, and released in 2020. It features 44 clinical and physical metrics from 541 women, of which 364 were healthy and 177 had a PCOS diagnosis. The target data is PCOS (positive = 1, negative = 0). The features can be grouped into two categories: Categorical Variable: Target, Pregnant(Y/N), hair growth(Y/N), Weight gain(Y/N), Hair loss(Y/N), Skin darkening(Y/N), Pimples(Y/N), Fast food(Y/N) etc. Numerical Vari-

able: Age (yrs), Height(Cm), RR(breaths/min), Hb(g/dl), Pulse rate(bpm), Weight (Kg), Cycle(R/I), Cycle length(days), No. of absorption, I beta-HCG(mIU/mL), Marriage Status(Yrs), PRL(ng/mL), PRG(ng/mL), Vit D3 (ng/mL) etc.

B. Data Preprocessing

Data preprocessing is a crucial stage in data mining, as the information intended for processing isn't always in optimal shape. In our selected dataset, Some attributes in the data samples had missing values, and some numeric data were saved as strings. So to avoid the troublesome issues, we handle the missing values by replacing them with the mean value and converting the object-type data into the numeric type. The dataset is unbalanced, with 32.7% of the sample having PCOS and 67.3% not. SMOTE (Synthetic Minority Over-sampling Technique) is employed to balance the class distribution by generating synthetic examples within the dataset. These synthetic samples are created through interpolation between existing minority-class samples, allowing algorithms to learn more generalized decision boundaries, which can lead to better classification performance. Using the `train_test_split()` and `cross_val_score()` methods from the scikit-learn library, the dataset was split into training and testing sets. Thirty percent of the data was set aside for testing, and seventy percent was used for training. Equalizing all feature measurements to a common scale is called feature scaling. Typically, if features are not scaled uniformly, the model tends to assign greater significance to higher values and less to lower ones, irrespective of their units. For better model efficiency and quicker optimization, this study utilizes standard feature scaling.

C. Visualization of Data

To gain understanding, identify trends, and clearly convey results, visualizing the data is a pivotal stage in the data analysis journey. The visualization of the correlation heatmap serves as a useful tool for quickly identifying which variables are most strongly associated with the target. In Figure 2, the correlation heatmap displays the relationships between features and the target variable, specifically highlighting those with a correlation coefficient exceeding 0.25. Cycle (R/I), AMH (ng/mL), weight gain (Y/N), hair growth (Y/N), skin darkening (Y/N), pimples (Y/N), fast food (Y/N), follicle no. (L), and follicle no. (R) are the most important features that are correlated with PCOS. This visualization helps with feature selection and model development.

D. Feature Selection Method

The process of feature selection is integral to machine learning and data analysis, wherein a subset of relevant features (variables or attributes) is chosen from the original feature set. The main objective is to improve the performance of a machine learning model by minimizing the number of dimensions, removing superfluous or irrelevant data, and emphasizing the most useful features. In this research, three distinct feature selection methods are employed.

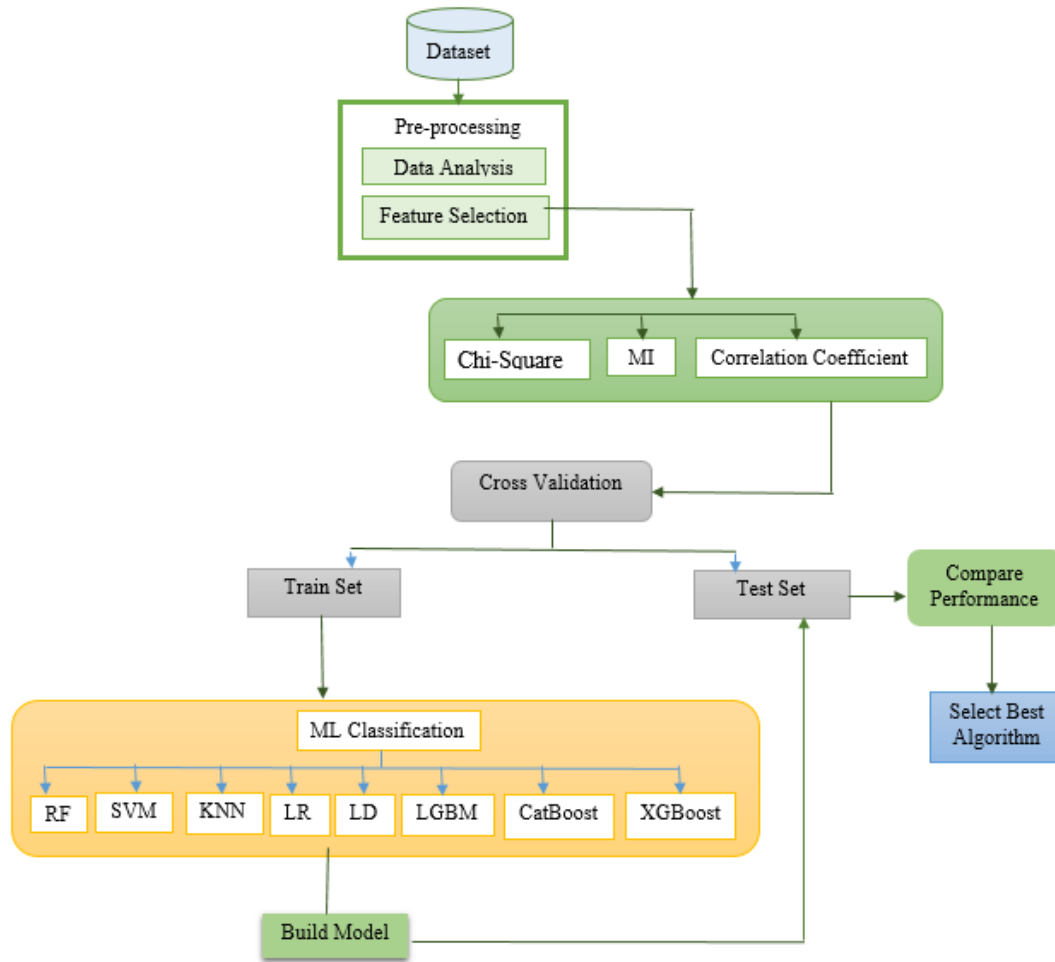


Fig. 1. Workflow Diagram

TABLE I
ACCURACY WITH AND WITHOUT CROSS-VALIDATION FOR EIGHT ALGORITHMS AND THREE FEATURE SELECTION METHODS

2*Classifier	Chi-Square (25)		Mutual Information (26)		Correlation Coefficient (20)	
	Test-train Split (%)	CV (%)	Test-train Split (%)	CV (%)	Test-train Split (%)	CV (%)
RF	87.73	93.11	88.34	92.51	86.5	92.5
SVM	91.33	90.74	91.41	90.95	89.57	89.76
KNN	84.04	86.02	86.50	85.64	84	86.82
LR	93.25	89.76	90.79	89.96	92.02	90.16
LD	92.63	90.55	90.79	90.35	90.18	90.55
LightGBM	87.73	93.51	88.34	90.76	85.88	91.94
CatBoost	88.34	93.90	88.34	92.33	86.5	91.93
XGBoost	87.73	90.56	87.73	90.96	85.27	91.34

1) *Chi Square*: The Chi-Square feature selection [8] technique is used, usually in a classification problem, to assess the independence of characteristics and the target variable. It ranks the features according to their Chi-Square values by calculating the Chi-Square statistic between each feature and the objective. Features with higher Chi-Square values are considered more relevant for classification.

2) *Mutual Information*: The Mutual Information (MI) feature selection [9] approach assesses each feature's dependence on the target variable. Higher MI scores indicate more significant features for prediction. It measures the amount of information about the target variable that can be obtained by understanding the feature.

3) *Pearson's Correlation*: The correlation coefficient, often Pearson's correlation, feature selection [10] method evaluates

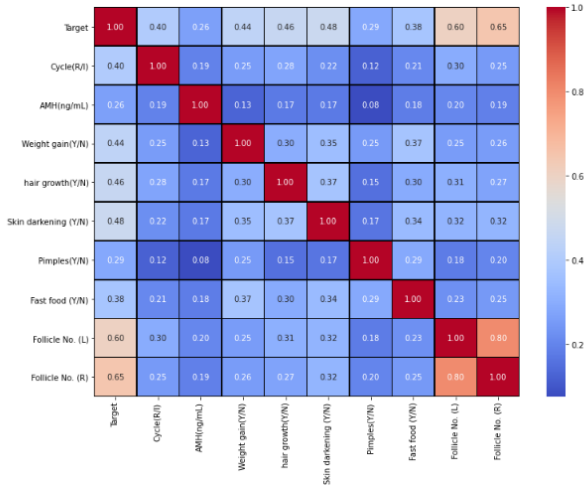


Fig. 2. Heatmap of Pearson correlation coefficient matrix

the linear correlation between each attribute and the target variable. In order to increase model accuracy and decrease dimensionality, it seeks to retain characteristics that exhibit a strong correlation with the goal and remove those that have little to no association.

E. Algorithms

In this section, we explore the machine learning methods used for forecasting PCOS. A brief overview of the operational mechanics of the machine learning models is provided. Our research scrutinizes eight different predictive machine learning models for their effectiveness in PCOS prediction.

1) *Random Forest*: The random forest (RF) [11] is an ensemble learning technique that aggregates multiple decision trees to enhance predictive performance and mitigate overfitting. It aggregates the outputs of individual trees through majority voting for classification tasks or averaging for regression tasks.

2) *Support Vector Machine*: The support vector machine (SVM) [12] is a supervised learning method frequently employed for both classification and regression tasks. It operates by identifying the hyperplane that effectively separates data

into distinct classes, maximizing the margin between these classes to enhance generalization.

3) *K-Nearest Neighbors*: The k-nearest neighbors (KNN) [13] is a simple yet powerful supervised learning technique used for classification and regression. It works by finding the k data points in the training set that are nearest to an input, and then predicting based on the average value or majority class among these neighbours.

4) *Logistic Regression*: Logistic regression (LR) [14] is a statistical method used for modeling binary outcomes, commonly employed in classification problems. By fitting data to a logistic curve, it calculates the likelihood that a given input falls into a particular category, aiding in decision-making in industries like marketing, finance, and healthcare.

5) *Linear Discriminant*: The linear discriminant (LD) [15] is a supervised classification technique that aims to find the linear combinations of features that best separate different classes. It is commonly used for dimensionality reduction and is effective in cases where the assumptions of equal class covariances and Gaussian distribution hold.

6) *Light Gradient Boosting Machine*: The Light Gradient Boosting Machine (LGBM) [16] is a high-performance, distributed, and efficient gradient-boosting framework specifically designed for speed and performance. It is widely used for various machine-learning tasks and can handle large datasets while providing high accuracy.

7) *CatBoost*: The CatBoost (Categorical Boosting) [17] is a gradient-boosting algorithm that excels at handling categorical variables without the need for prior encoding. Known for its robustness and high performance, it automatically deals with overfitting and offers interpretable model outputs.

8) *XGBoost*: The XGBoost (Extreme Gradient Boosting) [12] is a popular and powerful gradient-boosting algorithm known for its efficiency and high predictive accuracy. It optimizes the gradient boosting process through techniques like regularization and parallel computing, making it a top choice for various machine learning tasks, including classification and regression.

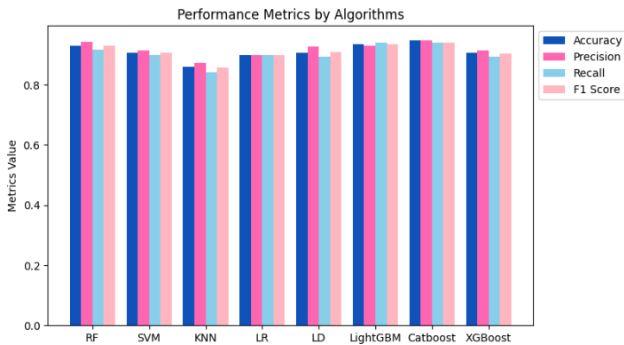


Fig. 3. Classification Report

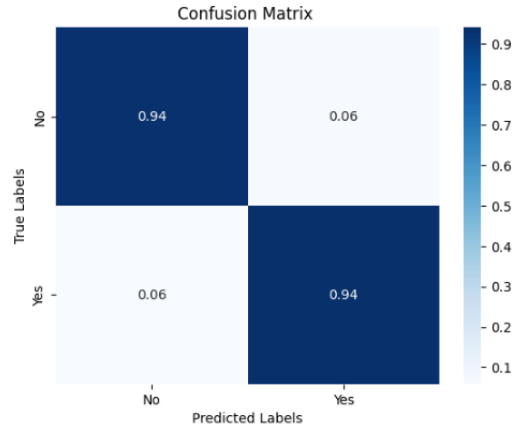


Fig. 4. Confusion Matrix

III. RESULT AND DISCUSSION

As mentioned previously, we have applied three feature selection methods as Chi-Square, Mutual Information, and Correlation Coefficient. 25, 26, and 20 significant features are selected subsequently for these three methods. Selected features are trained through SVM, Random Forest, KNN, Linear Discriminant Analysis, Logistic Regression, LightGBM, Catboost, and XGBoost model utilizing K-Fold cross-validation. In Table 1, we present the accuracy of the train-test split and cross-validation of the eight models for each feature selection technique. According to the table, using cross-validation improves the models' accuracy outcomes when compared to test-train split. Among the three feature selection methods, the result of the Chi-Square method was found to be better than the others. When we applied Chi-Square method and cross-validation, the accuracies obtained by different algorithms are: Random Forest 93.11%, SVM 90.74%, KNN 86.02%, LR 89.76%, LD 90.55%, LightGBM 93.51%, CatBoost 93.90%, and XGBoost 90.56%

In Figure 3, we have presented the accuracy, precision, recall, and F1 score of the algorithms with the important features selected by the Chi-Square method. We achieved better results for the CatBoost model, with an accuracy of 93.90%, precision of 93.72%, recall of 94.09%, and an F1 score of 90.43%, as calculated by the following equation.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 - Score = \frac{2 * TP}{2 * TP + FP + FN}$$

Figure 4 displays the confusion matrix for the CatBoost model using the Chi-Square feature selection strategy. The

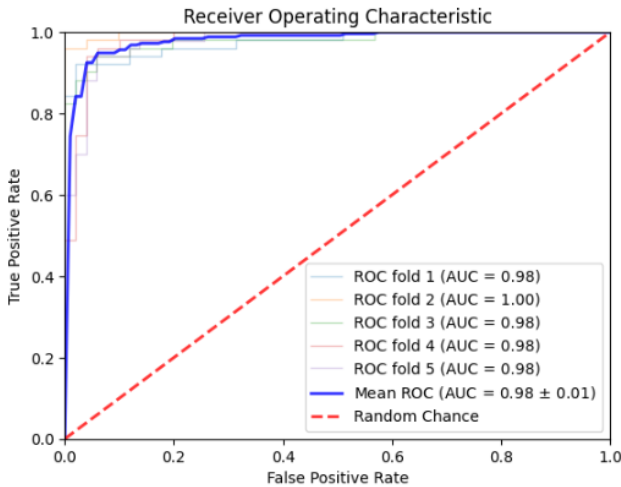


Fig. 5. ROC Curve

matrix presents the true positive (TP), true negative (TN), false positive (FP), and false negative (FN) values. From the matrix, we can say that both the TP and TN values are 0.94 and the FP and FN values are 0.06, which means that the model equally predicts both the positive and negative classes. In Figure 5, the ROC curve for the CatBoost classifier with the Chi-Square method is shown. This graph is generated using 5-fold stratified cross-validation, and it displays consistent predictive strength across subsets with a mean ROC value of 0.98, nearing the ideal of '1'. It suggests that the model's ability to categorize the target variable is both accurate and dependable.

With respect to the models and accuracy shown in Table 2, we contrast our findings with those of four previous studies on the same subject. When comparing the PCOS detection accuracy with other research, the highest accuracy is found in [4], [5], [6], and [7], at 91%, 89.02%, 90.9%, and 92.60%. Our best result is higher than these, which is 93.9%. Our model equally predicts both the positive and negative classes, and the performance of the models is improved compared to previous studies.

TABLE II
COMPARISON WITH PREVIOUS STUDY

Author	Ref.	Algorithm	Best Accuracy
Khoirun et al. [2023]	[4]	RF, LR	RF (91%)
Amsy et al.[2019]	[5]	NB, LR, KNN, CART, RF, SVM	RF (89.02%)
Vaidehi et al.[2021]	[6]	RF, SVM, LR, GNB, KNN	RF (90.9%)
Dana et al. [2022]	[7]	KNN, NN, NB, SVM, Tree, LR, LD	LD (92.60%)
This Study	—	RF, SVM, KNN, LR, LD, LightGBM, CatBoost, XGBoost	CatBoost (93.90%)

IV. CONCLUSION

Machine learning empowers computational systems to assimilate knowledge from data and arrive at insightful decisions. The diagnosis of PCOS often proves to be intricate due to the nebulous nature of its clinical symptoms. Consequently, this research aims to deliver rapid and effective diagnostic solutions for PCOS through the use of machine learning techniques. In order to improve the predictive accuracy of the models, we have assessed a range of machine learning algorithms using K-fold cross-validation, including SVM, Random Forest, KNN, Logistic Regression, Linear Discriminant Analysis, LightGBM, Catboost, and XGBoost. We also used feature selection techniques such as Chi-Square, Mutual Information, and Correlation Coefficient. Our research has effectively identified the most influential features through the utilization of the Chi-Square feature selection method, leading to a remarkable accuracy rate of 93.90% for the CatBoost algorithm. This study represents a substantial advancement over earlier efforts in the realm of PCOS detection. The enhanced predictive capacity of the model shows considerable promise

for healthcare applications, as the identified key features can serve as robust indicators for PCOS, thus facilitating early and swift risk assessment. Future endeavors should focus on the application of these findings in a clinical context, aiming to better healthcare results for women worldwide who are impacted by PCOS.

REFERENCES

- [1] E. A. Greenwood, L. A. Pasch, K. Shinkai, M. I. Cedars, and H. G. Huddleston, "Clinical course of depression symptoms and predictors of enduring depression risk in women with polycystic ovary syndrome: results of a longitudinal study," *Fertility and Sterility*, vol. 111, no. 1, pp. 147–156, 2019.
- [2] L. Reinhart, W. Knight, L. Roberts, and C. Mendes, "Hhs public access global burden of disease study 2010," *Lancet*, vol. 380, no. 9859, pp. 2163–2196, 2012.
- [3] A. Qayyum, J. Qadir, M. Bilal, and A. Al-Fuqaha, "Secure and robust machine learning for healthcare: A survey," *IEEE Reviews in Biomedical Engineering*, vol. 14, pp. 156–180, 2020.
- [4] K. Nisa, P. Purwono, B. P. Dewa, and S. K. Wibisono, "Model comparison of random forest and logistic regression algorithms in pcos disease detection," *Kesatria: Jurnal Penerapan Sistem Informasi (Komputer dan Manajemen)*, vol. 4, no. 1, pp. 73–79, 2023.
- [5] A. Denny, A. Raj, A. Ashok, C. M. Ram, and R. George, "i-hope: Detection and prediction system for polycystic ovary syndrome (pcos) using machine learning techniques," in *TENCON 2019-2019 IEEE Region 10 Conference (TENCON)*, pp. 673–678, IEEE, 2019.
- [6] V. Thakre, S. Vedpathak, K. Thakre, and S. Sonawani, "Pcicare: Pcos detection and prediction using machine learning algorithms," *Biosci Biotechnol Res Commun*, vol. 13, no. 14, pp. 240–244, 2020.
- [7] D. Hdaib, N. Almajali, H. Alquran, W. A. Mustafa, W. Al-Azzawi, and A. Alkhayyat, "Detection of polycystic ovary syndrome (pcos) using machine learning algorithms," in *2022 5th International Conference on Engineering Technology and its Applications (IICETA)*, pp. 532–536, IEEE, 2022.
- [8] Y. Zhai, W. Song, X. Liu, L. Liu, and X. Zhao, "A chi-square statistics based feature selection method in text classification," in *2018 IEEE 9th International conference on software engineering and service science (ICSESS)*, pp. 160–163, IEEE, 2018.
- [9] J. R. Vergara and P. A. Estévez, "A review of feature selection methods based on mutual information," *Neural computing and applications*, vol. 24, pp. 175–186, 2014.
- [10] H.-H. Hsu, C.-W. Hsieh, et al., "Feature selection via correlation coefficient clustering," *J. Softw.*, vol. 5, no. 12, pp. 1371–1377, 2010.
- [11] V. F. Rodriguez-Galiano, B. Ghimire, J. Rogan, M. Chica-Olmo, and J. P. Rigol-Sanchez, "An assessment of the effectiveness of a random forest classifier for land-cover classification," *ISPRS journal of photogrammetry and remote sensing*, vol. 67, pp. 93–104, 2012.
- [12] O. Jyoti, N. Islam, M. O. Faruq, M. A. I. Siddique, and M. H. Rahaman, "Autism spectrum disorder prognosis using machine learning algorithms: A comparative study," in *Intelligent Computing and Optimization: Proceedings of the 3rd International Conference on Intelligent Computing and Optimization 2020 (ICO 2020)*, pp. 754–765, Springer, 2021.
- [13] S. Vishwanathan and M. N. Murty, "Ssvm: a simple svm algorithm," in *Proceedings of the 2002 International Joint Conference on Neural Networks. IJCNN'02 (Cat. No. 02CH37290)*, vol. 3, pp. 2393–2398, IEEE, 2002.
- [14] K. Chaudhuri and C. Monteleoni, "Privacy-preserving logistic regression," *Advances in neural information processing systems*, vol. 21, 2008.
- [15] S. J. Prince and J. H. Elder, "Probabilistic linear discriminant analysis for inferences about identity," in *2007 IEEE 11th international conference on computer vision*, pp. 1–8, IEEE, 2007.
- [16] W. Liang, S. Luo, G. Zhao, and H. Wu, "Predicting hard rock pillar stability using gbd, xgboost, and lightgbm algorithms," *Mathematics*, vol. 8, no. 5, p. 765, 2020.
- [17] C. Zhang, X. Chen, S. Wang, J. Hu, C. Wang, and X. Liu, "Using catboost algorithm to identify middle-aged and elderly depression, national health and nutrition examination survey 2011–2018," *Psychiatry Research*, vol. 306, p. 114261, 2021.